Spread of infectious diseases with a latent period

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Abstract

Metapopulation Susceptible-Infected-Removed (SIR) model on a complex network is one of the models to investigate the epidemics of infectious diseases. We propose a modified the SIR model in which a latent period is included. Susceptible individuals turns to be in a pre-infected state, which gives a latent period, before the infected state. We call this model SIIR model. We discuss the difference in the spread of infectious diseases between the SIR model and SIIR model.

Keyword: epidemic, SIR model, metapopulation, complex network

1 Introduction

We are anxious about expansion of the epidemic area of Ebola hemorrhagic fever these days. The outbreak of infectious diseases is a problem which human has faced repeatedly from ancient times, for instance, the plague, Spanish flu, SARS, new strains of influenza, and so on. The epidemics of infectious diseases can be investigated from the viewpoint of a complex network. Well known Susceptible-Infected-Removed (SIR) model on a complex network connecting individuals has been extended to the SIR model on a metapopulation network [1, 2, 3]. In this paper, we propose a modified SIR model in which a latent period is included.

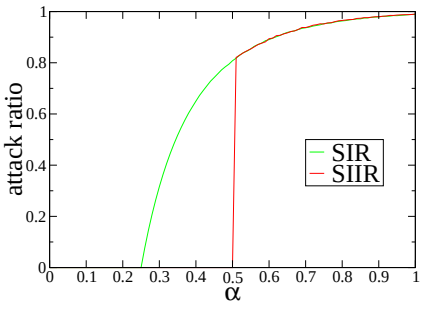

Figure1: Attack ratio as a function of the infection rate α. The recovery rate is $\beta = 0.5$ for the SIIR model and $\beta = 0.25$ for the SIR model.

2 Model

In the metapopulation SIR model, susceptible individuals turn to be infected at a rate proportional to the number of infected individuals, and all individuals move to the neighboring population at a constant rate. However, in the real world, infected and sick people would not move as frequently as healthy ones do. Thus, we introduce a latent period to the SIR model. Susceptible individuals turns to be in a pre-infected state, which gives a latent period, before the infected state. We assume that susceptible, pre-infected and recovered individuals move to the neighboring population at a constant rate but infected ones do not. We call this model SIIR model.

3 Results

In this paper, we discuss the difference in the spread of infectious diseases between the SIR model and the SIIR model. In both the models, infectious diseases globally spread when the infection rate is larger than the recovery rate. However, as shown in Fig. 1, the attack ratio is discontinuous in the SIIR model, while it is continuous in the SIR model. We will present a more detailed discussion in this colloquium.

References

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